

SAVITRIBAI PHULE PUNE UNIVERSITY

A Project Report On
VitalX – Healthcare Record Storage on Blockchain

Submitted Towards The
Partial Fulfillment Of The Requirements Of
BSc in Blockchain Technology

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Abstract

The healthcare system today faces significant challenges in securely managing medical records. Patients often lose access to their medical history when moving between hospitals or cities. Existing centralized systems are prone to data breaches and lack transparency. To overcome these issues, **VitalX** proposes a decentralized healthcare record storage system built on the Ethereum blockchain. Using IPFS for secure document storage and smart contracts for verification, the system ensures tamper-proof and transparent data handling. Patients have complete ownership of their records and can control who can access or verify them. This project demonstrates how blockchain can revolutionize healthcare data management by ensuring privacy, integrity, and accessibility through cryptographic verification and smart contracts.

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Chapter 1: Synopsis

Project Title: VitalX – Healthcare Record Storage on Blockchain

Problem Statement: Patients often lose or mismanage their medical data while switching healthcare providers. Existing centralized systems are insecure and inefficient. There is a need for a decentralized and tamper-proof system that ensures secure medical data storage and patient control.

Solution: The proposed DApp leverages blockchain and IPFS to securely store healthcare data. Doctors verify prescriptions and vaccination certificates before being marked as valid on the blockchain. Patients retain ownership of their records and can grant or revoke access securely.

Objectives: To ensure patient data ownership and control, to use blockchain and IPFS for tamper-proof health record storage, to enable doctor verification through smart contracts, and to maintain transparency, privacy, and immutability of records.

Chapter 2: Technical Keywords

1. Blockchain
2. Smart Contracts
3. IPFS
4. Ethereum
5. WalletConnect
6. AES/RSA Encryption
7. Decentralization
8. Data Privacy
9. Verification System
10. React.js

Chapter 3: Introduction

The **VitalX** project introduces a decentralized approach to healthcare data storage and management. Traditional healthcare systems suffer from inefficiency, lack of transparency, and data tampering risks. This project uses Ethereum smart contracts for verification, IPFS for decentralized file storage, and AES/RSA encryption to protect patient privacy. Patients can access their medical data globally, grant or revoke doctor access, and maintain complete control over their information.

Chapter 4: System Design Overview

Modules:

Patient Module – Upload and manage medical records.

Doctor Module – Verify and sign uploaded medical documents.

Admin Module – Onboard doctors and manage platform.

Smart Contract – Handle metadata, verification, and access logic.

Workflow:

1. Patient uploads encrypted health record to IPFS.
2. Metadata (IPFS hash, timestamp, verifier ID) stored on Ethereum.
3. Doctor verifies the record through smart contract.
4. Patient can view, share, or revoke access anytime.

Chapter 5: Tools and Technologies

Frontend: React.js

Blockchain: Ethereum (Sepolia Testnet)

Storage: IPFS

Smart Contract: Solidity

Testing: Hardhat

Encryption: AES for data, RSA/ECIES for key sharing

Authentication: WalletConnect

Database (optional): MongoDB for caching logs

Chapter 6: Testing and Results

The system was tested using Hardhat and Ganache for contract deployment and verification. Each module was validated for security, performance, and functionality.

Smart contract deployed successfully on Sepolia testnet.

IPFS integration verified – records uploaded securely.

Doctor verification and access control functioning correctly.

WalletConnect authentication verified on test accounts.

Chapter 7: Conclusion and Future Scope

The **VitalX** project provides a secure and transparent healthcare record management system using blockchain technology. By decentralizing medical record storage and allowing patient ownership, VitalX ensures data privacy, accessibility, and authenticity. Future improvements include integrating AI-based health analytics, mobile app support, and interoperability with government health systems.